

Apply to be a Science Corps Fellow - Entry #3372

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Gender
Female
Residence: City (State), Country
Fort Collins (Colorado), USA
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United States of America
When did you or do you expect to complete your PhD?
July 2024
Please tell us briefly about your university education (institution, city, country, year)
Colorado State University is a state-funded university located in Fort Collins, Colorado, USA. CSU is a large R1 university that is strong in the sciences and well-known for its veterinary sciences program. It also houses teaching-centered institutions such as the Institute for Teaching and Learning (TILT) which provides education on research-based strategies for academic success. I attended CSU for my Bachelor of Science degree from 2005-2008, Master of Science degree from 2017-2019, and my PhD from 2019-2024. While I have made my home in Fort Collins and enjoy it immensely, I am ready and excited to live abroad.
Why would you like to be a Science Corps Fellow?
I have always wanted to serve internationally and considered joining the Peace Corps in my early 20s. I learned about Science Corps the first month starting my MS degree, and Science Corps has been on my radar ever since. As I was nearing the completion of my PhD, I gave myself a calendar reminder to look for the open application, which I'm now seeing for the first time since graduating. I'm excited to be applying. I want to teach and inspire students to feel confident in their own practice of science, and I especially want to do this in international communities that may not have access to the science I have been so fortunate to be immersed in for over a decade. Science is a skill and a practice that we all do in our daily lives, often without recognizing it. When we garden,

exercise, cook, decide what to purchase, how to tend a wound, decide if the milk in the fridge is sour, choose when to go to bed—every one of these experiences is a scientific endeavor. It is, of course, the microbes in the milk that devour the sugars and excrete foul-smelling products that tells us that we should probably have an egg for breakfast instead. Every time we observe, reflect, and make a decision, we are enacting the essence of science. I want to share the joy that comes from practicing science in daily life.

I am a life-long learner and a natural leader. While I can find myself enthralled to learn just about anything, I have been particularly drawn to molecular biology. At the scale of the molecule, we can't see anything that is happening. Atoms are mere nanometers in width, and we need a microscope to see things even 100 times that size. But we have methods to identify and test the invisible workings of life at the nanoscale, and this has enchanted me to study it for over 15 years. Now with a PhD, I am looking to share not only the foundation of knowledge I've gained, but more importantly, the process of exploring a question, developing an elegant experiment, and discovering an answer.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

During my PhD, I focused on education by teaching an Introduction to Microbiology course as a Visting professor at Colorado College. I enjoyed standing at the front of a classroom asking, "How do we know what we know?" While I was busy designing experiments and generating data for my thesis, I enjoy mentoring and opted to mentor nine undergraduate research students by bringing them into the lab and providing one-on-one training in actual experimentation. Students ranged from freshman-seniors and included an international student from Ecuador.

I am adventurous and have traveled to ~30 countries on almost every continent barring South America and Antarctica. I have yet to live in another country, thus I would love nothing more than for the opportunity to teach science internationally.

In addition to travel, I have lived and worked out of a tent or out of the back of my truck for months at a time. I first lived out of a tent when I joined a trail crew to improve trails to high mountain peaks (up to 14,000 ft. (~4200 m.) for three months. I've since lived out of the back of my truck while working in conservation biology—we patrolled for nesting sea turtles on the Texas coast and scoured the deserts of Southern California for Desert Tortoise. I am skilled at developing systems to manage less-than-ideal weather conditions and have experience keeping groups calm and focused in likewise challenging situations.

I have worked in many fields of science, studying everything from biofuels to rare plants to industrial beer production. I am particularly well versed in biochemistry, but my interests and experience cover: gardening, farming, agriculture, barley breeding, plant pathology, plant identification, rare plant surveys, desert tortoise surveys, sea turtle surveys, conservation field biology, rice genetics, microbial genetics, microbial metabolism, protein purification, biofuel production, DNA and RNA sequence analysis, genetic engineering, synthetic protein design, etc. I love science and hope to share this passion by working with Science Corps.

Where do you imagine yourself professionally in the future? Check all that apply.

In science education

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

I am currently looking to teach at the college level. If I am not accepted to join Science Corps, I am applying to teach microbiology at Front Range Community College in the Spring of 2026.

In 10 years, I would ideally like to see myself teaching and communicating science to a broad audience. This could look like teaching at a primarily undergraduate university, outreach such as leading a science-based service organization, or running a science communication program. I love science, but I love talking about science even more.

Please upload your CV (.pdf)

 [CV_SWilliams.pdf](#)

How did you find out about Science Corps?

Referred by a friend/colleague/supervisor in academia

[Science Corps](#)